

Alp Sahin

Department of Mechanical Engineering and Mechanics, Lehigh University
Building C, 113 Research Drive, Bethlehem, PA 18015
+1 (508) 310-3081 • as.alpsahin@gmail.com • [linkedin.com/in/sahinalp/](https://www.linkedin.com/in/sahinalp/) • github.com/asahin1

EDUCATION

Lehigh University

Ph.D. in Mechanical Engineering and Mechanics

Bethlehem, PA

Aug. 2021 – Expected: December 2026

Worcester Polytechnic Institute

M.S. in Robotics Engineering

Worcester, MA

Aug. 2019 – May 2021

Bogazici University

B.S. in Mechanical Engineering

Istanbul, Turkey

Sept. 2014 – June 2019

RESEARCH INTERESTS

My research philosophy centers on translating complex robotics problems into mathematical programs and developing algorithms to solve them efficiently. I am interested in combinatorial problems that arise in planning for robotic manipulation, as well as multi-robot and tethered-robot systems. I investigate how ideas from graph theory, algebraic topology, and mixed-integer programming can be leveraged to develop search- and optimization-based approaches.

RESEARCH EXPERIENCE

Autonomous and Intelligent Robotics Laboratory

Research Assistant

August 2021 – Present

Lehigh University

Advisor: Dr. Subhrajit Bhattacharya

- Developed search- and optimization-based planning algorithms for multi-robot and tethered-robot systems.
- Implemented optimization algorithms for kinematic control of snake-like robots and control of networked systems.

Manipulation and Environmental Robotics Lab

Graduate Researcher

January 2020 – May 2021

Worcester Polytechnic Institute

Advisor: Dr. Berk Calli

- Formulated motion primitives for an in-hand manipulation setup involving variable friction fingers and an articulated arm.
- Developed a search-based planning algorithm for achieving target contact configurations on objects.

Robotics, Mobility and Cyberphysical Systems Lab

Graduate Researcher

May 2020 – June 2021

Worcester Polytechnic Institute

Advisor: Dr. Xiangrui Zeng

- Developed an RL-based feature selection algorithm for constrained data collection, maximizing the performance of supervised-learning models trained on delayed data streams.
- Created a benchmarking tool to evaluate the performance of dynamic data collection policies.

PUBLICATIONS

Journal

- [J1] **A. Sahin** and S. Bhattacharya, “Topo-Geometrically Distinct Path Computation Using Neighborhood-Augmented Graph, and Its Application to Path Planning for a Tethered Robot in 3-D,” *IEEE Transactions on Robotics*, 2025. [\[PDF\]](#)[\[Code\]](#)
- [J2] **A. Sahin**, N. Kozachuk, R. S. Blum, and S. Bhattacharya, “Spectrum optimization of dynamic networks for reduction of vulnerability against adversarial resonance attacks,” *Frontiers in Complex Systems*, 2025. [\[PDF\]](#)
- [J3] Z. Wang, **A. Sahin**, and X. Zeng, “Efficient Data Collection for Connected Vehicles With Embedded Feedback-Based Dynamic Feature Selection,” *IEEE Transactions on Intelligent Vehicles*, 2024. [\[PDF\]](#)
- [J4] X. Wang, **A. Sahin**, and S. Bhattacharya, “Coordination-free Multi-robot Path Planning for Congestion Reduction Using Topological Reasoning,” *Journal of Intelligent & Robotic Systems*, 2023. [\[PDF\]](#)

Conference

- [C1] **A. Sahin**, S. Brawner, M. Bilsky, and S. Bhattacharya, “Real-Time Joystick-based Kinematics Control of a Snake-like Robot with Robustness Near Singular Configurations,” IEEE Conference on Control Technology and Applications (CCTA), 2024. [PDF]
- [C2] **A. Sahin**, A. J. Spiers, and B. Calli, “Region-Based Planning for 3D Within-Hand-Manipulation via Variable Friction Robot Fingers and Extrinsic Contacts,” IEEE International Conference on Robotics and Automation (ICRA), 2021. [PDF][Code-Planner][Code-Simulation]
- [C3] **A. Sahin** and X. Zeng, “Feedback-Based Dynamic Feature Selection for Constrained Continuous Data Acquisition,” American Control Conference (ACC), 2021. [PDF]
- [C4] **A. Sahin** and X. Zeng, “A Framework for Benchmarking Feedback-Based Dynamic Data Collection Methods in Connected Vehicle Networks,” SAE Technical Paper, 2021.

TEACHING EXPERIENCE

Department of Mechanical Engineering and Mechanics

Graduate Teaching Assistant

January 2022 - May 2023

Lehigh University

Courses: ME 240 – Manufacturing (Spring 2022, Spring 2023), *with Dr. David C. Angstadt*

AWARDS

Lehigh University Rossin College Doctoral Student Fellow

August 2021

2nd Place in WPI Graduate Research Innovation Exchange

April 2021

SERVICE

Scholarly Reviews

IEEE Robotics and Automation Letters (RA-L), IEEE Transactions on Intelligent Vehicles (T-IV), IEEE Transactions on Robotics (T-RO), IEEE International Conference on Robotics and Automation (ICRA), IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)

Outreach

Robotics activity organizer and speaker at Lehigh CHOICES Summer Camp

June 2022 and June 2023

RELEVANT PROJECTS

Design and Development of an Optical Force Sensor for an Extensible Colonoscopy Robot

- Designed the optical circuit using laser diode, light sensor, and fiber optic cables
- Designed and assembled CAD parts for the sensor mechanism
- Conducted experiments with the optical setup
- Tuned and tested the force sensor

Cost Sensitive Sensor Selection for State Estimation Using Kalman Filter

- Used Unscented Kalman Filter (UKF) for position estimation of an agent on 2D grid
- Implemented simulation based value iteration and dynamic programming (DP) on a finite horizon problem
- Developed a sensor selection algorithm based on the DP solution

Design and Simulation of Hybrid Position-Force Controller for 7-DOF Kuka Arm

- Designed a controller that combines impedance and PD controllers to achieve a compliant behavior at the end effector while maintaining high motion tracking accuracy
- Tested the controllers on a planar writing task in simulation using Asynchronous Multi-Body Framework (AMBF)

Assistive Rehabilitation System for Patients Undergoing Physiotherapy Treatment

- Modeled a human arm using Blender
- Derived forward and inverse kinematics for both human and Kuka arm models
- Used Rigid Body Dynamics Library (RBDL) to derive the dynamics of the Kuka arm
- Designed an inverse dynamics controller for trajectory tracking tasks

- Simulated the models in Asynchronous Multi-Body Framework (AMBF)

Autonomous Mobile Robot Navigation and Obstacle Avoidance

- Implemented artificial potential field for path planning and obstacle avoidance
- Tested the motion plan in simulation using V-REP
- Used ROS nodes to communicate the motion plan to actual robots
- Demonstrated the path planning and obstacle avoidance on mobile robots equipped with stereo cameras

SKILLS

Programming and Development: C/C++, Python, MATLAB, Git, CMake, Docker, Bash

Software: ROS, Gazebo, Gurobi, OpenCV, Drake, Mathematica, CVX, OMPL

Typesetting & Editing: L^AT_EX, Inkscape

PROFESSIONAL EXPERIENCE

Beycelik GESTAMP Teknoloji ve Kalip A.S.

Tooling Process and Design - Intern

Bursa, Turkey

January 2019 - February 2019

TOFAS Turk Otomobil Fabrikasi A.S.

Press and Die Production - Intern

Bursa, Turkey

June 2018 - July 2018

GESTAMP Metalbages

Process Engineering - Intern

Santpedor, Spain

July 2017 - August 2017